Chemistry 3A – Fall 2026 Name _____

Sections 25810-25811 10-11 September 2026

Special Midterm Examination #1 ID _____

Select the **BEST** response for the question. Point values: multiple choice 3 pt; true/false 1 pt

1. Which number below is standard (decimal) notation for 2.350×10^2?
   a) 235 b) 235.0 c) 23.50 d) 0.02350

2. Of the orders shown below from largest to smallest size (atomic radius), which show the correct order?
   a) F > S > Mg > Cs b) Cs > S > Mg > F c) Cs > Mg > S > F d) Mg > Cs > F > S

3. What is the correct electronic configuration for element **sulfur**?
   a) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$ b) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$ c) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$ d) $1s^2\ 2s^2\ 2p^6\ 3s^1\ 3p^4$

4. Which element has five (5) valence electrons?
   a) Nitrogen (N) b) Boron (B) c) Carbon (C) d) Oxygen (O)

5. The elements in Group 2 (old designation 2A) are known by what name?
   a) Halogens b) Transition Metals c) Alkali Metals d) Alkaline Earth Metals

6. Which of these ions has the same electron configuration as **neon** (Ne)?
   a) O^{2-} b) Na^- c) F^+ d) Mg^{2-}

7. Using the electron filling rules (Aufbau principle), how many unpaired electrons are present in a neutral atom of the element **titanium** (**Ti**)?
   a) 4 b) 2 c) 1 d) 0

8. What is the result of the multiplication operation with correct significant digits: 20.6807×6.91
   a) 142.904 b) 143 c) 142.9 d) 142.9036

9. What is the density of a sample of metal shot with a mass of 12.6 g placed inside a graduated cylinder containing water, if the volume reading is 15.0 mL before adding the metal and 19.5 mL after? *(All choices have units of g/mL)*
   a) 2.8 b) 0.65 c) 0.36 d) 56.7

10. Consider the following atomic symbols, where X element:
    I. $^{42}_{20}X$ II. $^{46}_{22}X$ III. $^{44}_{20}X$ IV. $^{44}_{21}X$
    Which pairs are isotopes of the same element?
    a) I and IV b) II and IV c) I and III d) III and IV

11. Which subatomic particle(s) are located ***inside*** the nucleus of an atom?
    a) electrons only b) protons only c) protons and neutrons d) protons and electrons

12. Which statement is a postulate of Dalton's atomic theory?
    a) Atoms are composed of smaller subatomic particles called protons, neutrons, and electrons.
    b) Electrons travel around the nucleus in specific circular energy levels.
    c) Chemical reactions involve the rearrangement of atoms, not their creation or destruction.
    d) Atoms of different elements have identical masses and chemical properties.

13. A student measures a volume of **0.00380 liters**. How should this quantity be expressed in scientific notation?
    a) 3.8×10^{-3} L b) 3.80×10^3 L c) 3.8×10^3 L d) 3.80×10^{-3} L

14. Which principle states that no two electrons in an atom can have the exact same set of four quantum numbers (meaning an orbital can hold a maximum of two electrons with opposite spins)?
a) Hund's rule b) Pauli exclusion principle c) Aufbau principle d) Dalton's rule

15. If a chemical reaction takes place at $25^{\circ}\mathrm{C}$, what is the equivalent temperature on Kelvin scale?
a) 248 K b) 298 K c) –248 K d) 571 K

16. Which component of the scientific method represents a tentative, testable explanation for a set of observations before it has been thoroughly tested?
a) hypothesis b) (scientific) law c) theory d) experiment

17. Perform the calculation and apply decimal-place rules: 18.452 – 0.12 + 4.3051
a) 22.637 b) 22.6 c) 22.64 d) 22.6371

18. Which of the following elements is classified as a **metalloid (semimetal)** on the Periodic Table?
a) silicon b) aluminum c) phosphorus d) calcium

19. In Bohr's model of the atom, what happens when an electron absorbs a photon of energy and transitions to a higher energy level?
a) emit a photon of visible light
b) it escapes the atom completely to turn the atom into an ion
c) it moves from a lower energy level to an excited, higher energy level
d) it alters the number of protons, changing the atomic number (Z)

20. Perform the calculation and apply significant-digit rules: 0.0450 ÷ 0.25
a) 0.18 b) 0.180 c) 0.2 d) 0.1800

21. Which of the following statements about **protons** is correct?
a) They orbit the nucleus in the electron cloud outside
b) They carry a negative electrical charge
c) The sum of protons and electrons equals the mass number (A)
d) They have a relative mass of approximately 1 amu

22. How many individual orbitals are present in a $\boldsymbol{d}$ **subshell**?
a) 1 b) 3 c) 5 d) 7

23. If the mass of an object is divided by its density (Mass/Density), which of the following units would be most appropriate for the resulting quantity?
a) grams (g) b) milliliters (mL) c) grams per milliliters (g/mL) d) centimeters (cm)

24. How many neutrons are present in an atom of the **carbon-14** isotope ($Z = 6$)?
a) 8 b) 6 c) 20 d) 14

25. In atomic physics, electron energy levels inside an atom are described as **quantized**. What does this term imply about an electron moving between energy levels?
a) It can gain or lose any arbitrary amount of energy along a continuous scale.
b) Its charge changes from negative to neutral or positive during the transition.
c) It temporarily converts its mass into pure light energy during movement.
d) It can only absorb or emit specific discrete amounts of energy corresponding to the difference between levels.

Continue to next page for True/False questions

Select (a) if TRUE and (b) if FALSE for these ten true-false questions

26. The atomic mass of an element reported on the Periodic Table represents the weighted average mass of all its naturally occurring isotopes.
    a) true b) false

27. In the scientific method, formulating a hypothesis always occurs prior to making any initial observations.
    a) true b) false

28. Ionization energy is defined as the minimum amount of energy required to remove an electron from a neutral gaseous atom.
    a) true b) false

29. Two atoms with different mass numbers (A) but the same atomic number (Z) belong to different chemical elements
    a) true b) false

30. Neutrons have a significantly smaller mass than protons.
    a) true b) false

31. In the measurement 0.020440 g, the leading zeros are significant, but the trailing zero is not
    a) true b) false

32. In standard and customary scientific notation ($N \times 10^n$), the coefficient N must be a number greater than or equal to 1 and strictly less than 10.
    a) true b) false

33. When the single electron in a hydrogen atom absorbs energy and transitions from $n = 1$ to $n = 2$, the atom is said to be in an "excited state"
    a) true b) false

34. Absolute zero (0 K) corresponds to the temperature at which water boils and turns into steam.
    a) true b) false

35. According to the Aufbau principle, electrons fill the $4s$ subshell before beginning to occupy the lower-energy $3d$ subshell
    a) true b) false

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